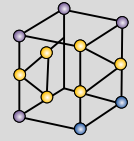


Input

Crystal Structure



$X = (F, R, L)$

Atomic features

$$F = \{f_s\} \in \mathbb{R}^{N \times d_f}$$

Coordinates

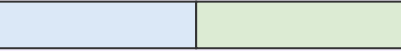
$$R = \{r_s\} \in \mathbb{R}^{N \times a}$$

Lattice parameters

$$L = (a, b, c, \alpha, \beta, \gamma)$$

Dual-channel Embedding

d/2 (XY Channel) d/2 (Z Channel)



Separate encoding for XY (in-plane) and Z (out-of-plane) channels

Token Sequence (Concat)

$$x \in \mathbb{R}^{(N_{\text{atom}}+3) \times d}$$



Atom tokens

Lattice tokens

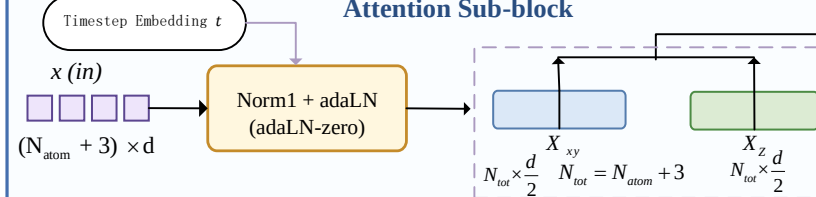
N_{atom}

3

+ sin-cos Positional Encoding

Unified Block (depth = 18)

Attention Sub-block



Self-Attention (XY) + RelPosBias

Multi-Head Self-Attention

Self-Attention (Z)

Multi-Head Self-Attention

Cross-Attention (Main)

$$Q_z \rightarrow K, V_{xy}$$

$$Gate_{z \rightarrow xy} \text{ (init } \approx 0.78)$$

Cross-Attention (Auxiliary)

$$Q_{xy} \rightarrow K, V_z$$

$$Gate_{xy \rightarrow z} \text{ (init } \approx 0.10)$$

$$Out_z$$

$$Out_{xy}$$

$$\oplus$$

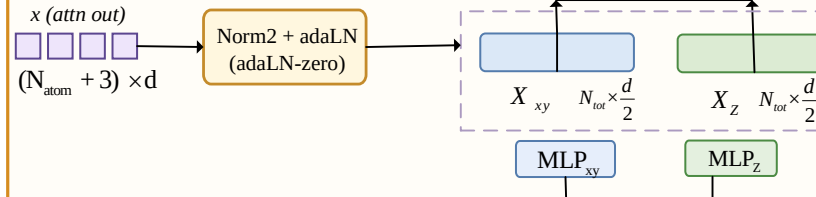
$$\oplus$$

$$\text{Concat}(out_{xy}, out_z) \rightarrow \text{Linear (proj)}$$

$$\times Gate_{msa} \text{ (adaLN)}$$

$$x \text{ (attn-out)}$$

MLP Sub-block



$$MLP_{xy}$$

$$MLP_z$$

$$\text{Concat}(\cdot) \rightarrow \text{Fusion}$$

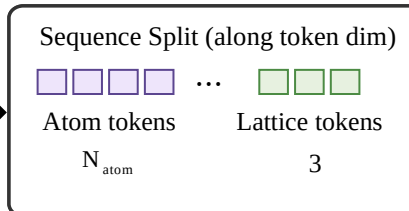
$$\text{Linear} - \text{GELU} - \text{Linear} \times Gate_{mlp}$$

$$\times Gate_{mlp} \text{ (adaLN)}$$

$$x \text{ (block out)}$$

— Shared Path(train + inference) - - - Backprop (train only)

- - - Training Supervision Only — modulates all adaLN modules within the block.



Atom Final Layer

adaLN-LN + Linear

(d → 5)

$Atom_{out}$

Lattice Final Layer

adaLN-LN + Linear

(d → 3)

$Lattice_{out}$

$$Atom_{pred} \in \mathbb{R}^{N_{\text{atom}} \times 5}$$

$$Lattice_{pred} \in \mathbb{R}^{3 \times 3}$$

$$\oplus$$

$$\oplus$$

$$t=0?$$

$$\text{Decoder}$$

$$\text{Generated Crystal Structure}$$

$$(\hat{F}, \hat{R}, \hat{L})$$

$$\mathcal{L}_{\text{base}}$$

$$\mathcal{L}_{\text{phys}}$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{base}} + \mathcal{L}_{\text{phys}}$$

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